

LONGEST STAY DETECTION.

PRESENTED BY : WISIT SUWANNAO (วิศิษฐ์ สุวรรณนาวี)



INTRODUCE MYSELF



WISIT SUWANNAO

Super AI Engineer
Season 6 Level 2

EXPERIENCE

- **AI Engineer Intern** at Oneput Technology Company Limited
- **Winner**, Super AI Engineer SS6 Level 2 - Demand Forecasting Coffee Chain Hackathon
- **Top 10 Finalists** - CAI CAMP 2025: PromoAutomate
- **Former participants** in Super AI Engineer Season 5 Level 1, Level 2(Online)



LinkedIn



GitHub

AGENDA

Problem Statement and Constraints

Solution of the Problem

- Definition of Stationary

- End-to-End Pipeline

- Robustness Decisions

- Result Preview

- Final Answer

Deliverables and Repository

Limitations and Next Steps

PROBLEM STATEMENT AND CONSTRAINTS

PROBLEM STATEMENT AND CONSTRAINTS

- **Input:** video raw footage from entrance.mov
- Detect the person who remained stationary the longest
- Report track id, duration, start time and end time
- Read FPS and duration from OpenCV metadata
- **Output:** source code, README, requirements, annotated video and short idea description



entrance.mov
raw video footage

SOLUTION OF THE PROBLEM

SOLUTION OF THE PROBLEM

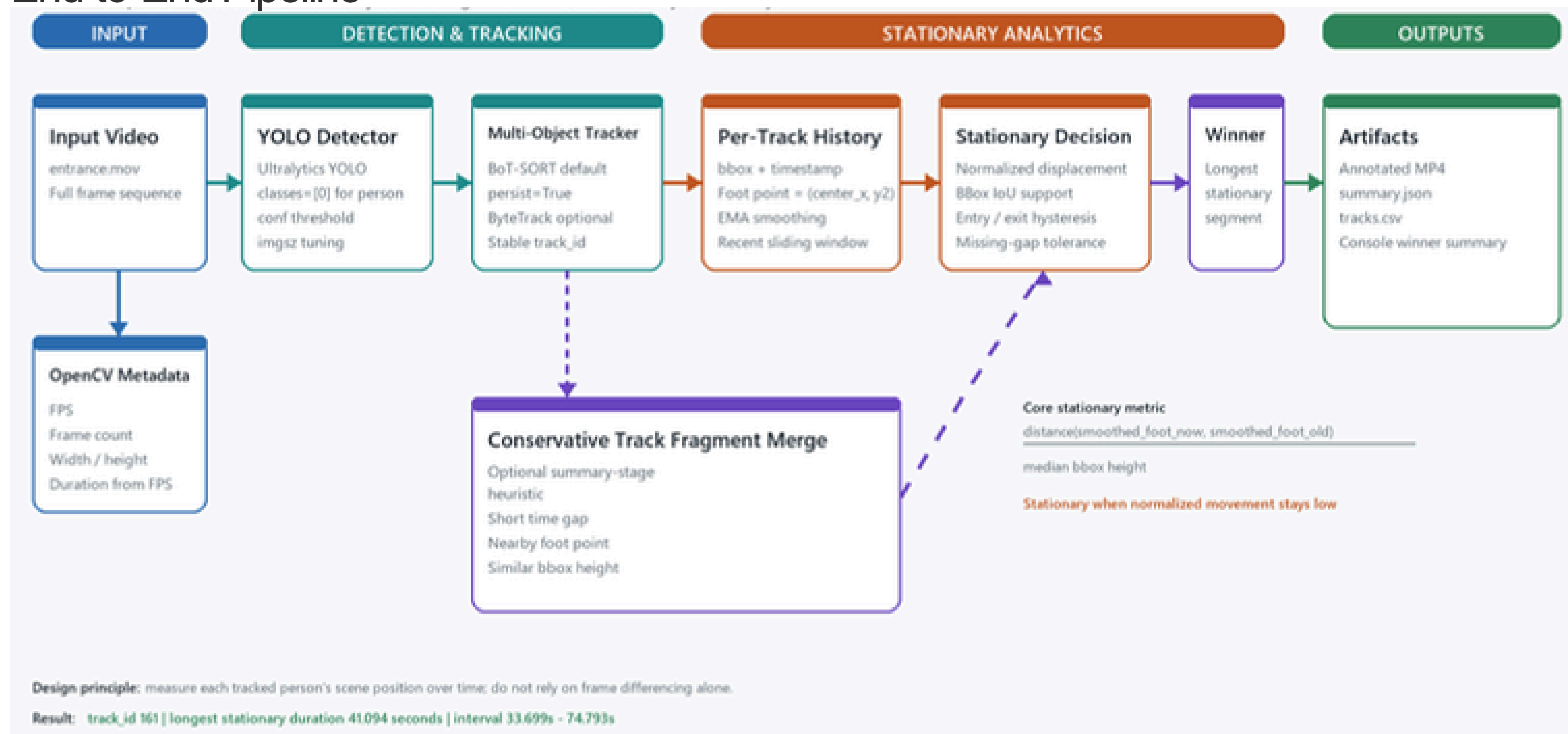
Definition of Stationary

Stationary = scene position is almost unchanged

- **foot_point**: ใช้ (center_x, y2) แทนตำแหน่งคนบนพื้น
 - **norm_disp** = $\text{distance}(\text{current_smoothed_foot}, \text{old_smoothed_foot}) / \text{median_bbox_height}$
- ใช้แทนระยะการเคลื่อนที่ด้วย bbox height เพื่อให้ threshold ใช้ได้ดีขึ้นกับทั้งคนใกล้และไกลกล้องครับ

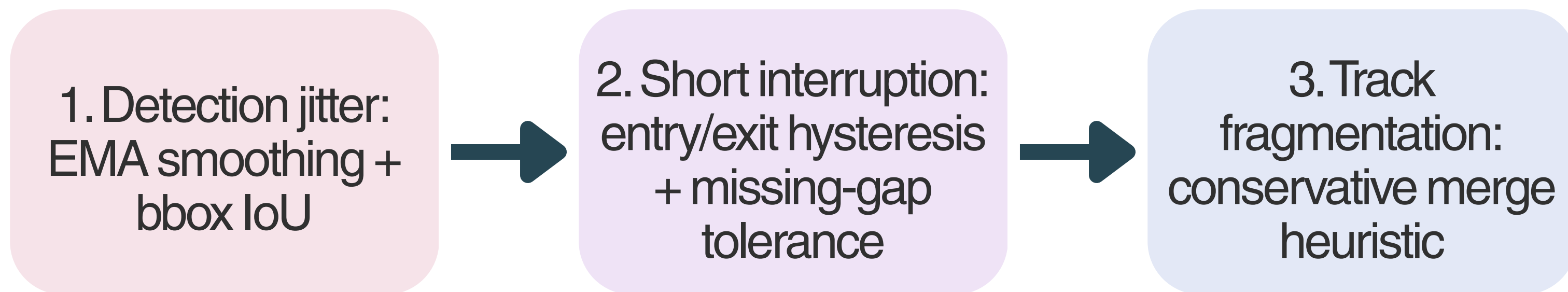
SOLUTION OF THE PROBLEM

End-to-End Pipeline



SOLUTION OF THE PROBLEM

Robustness Decisions



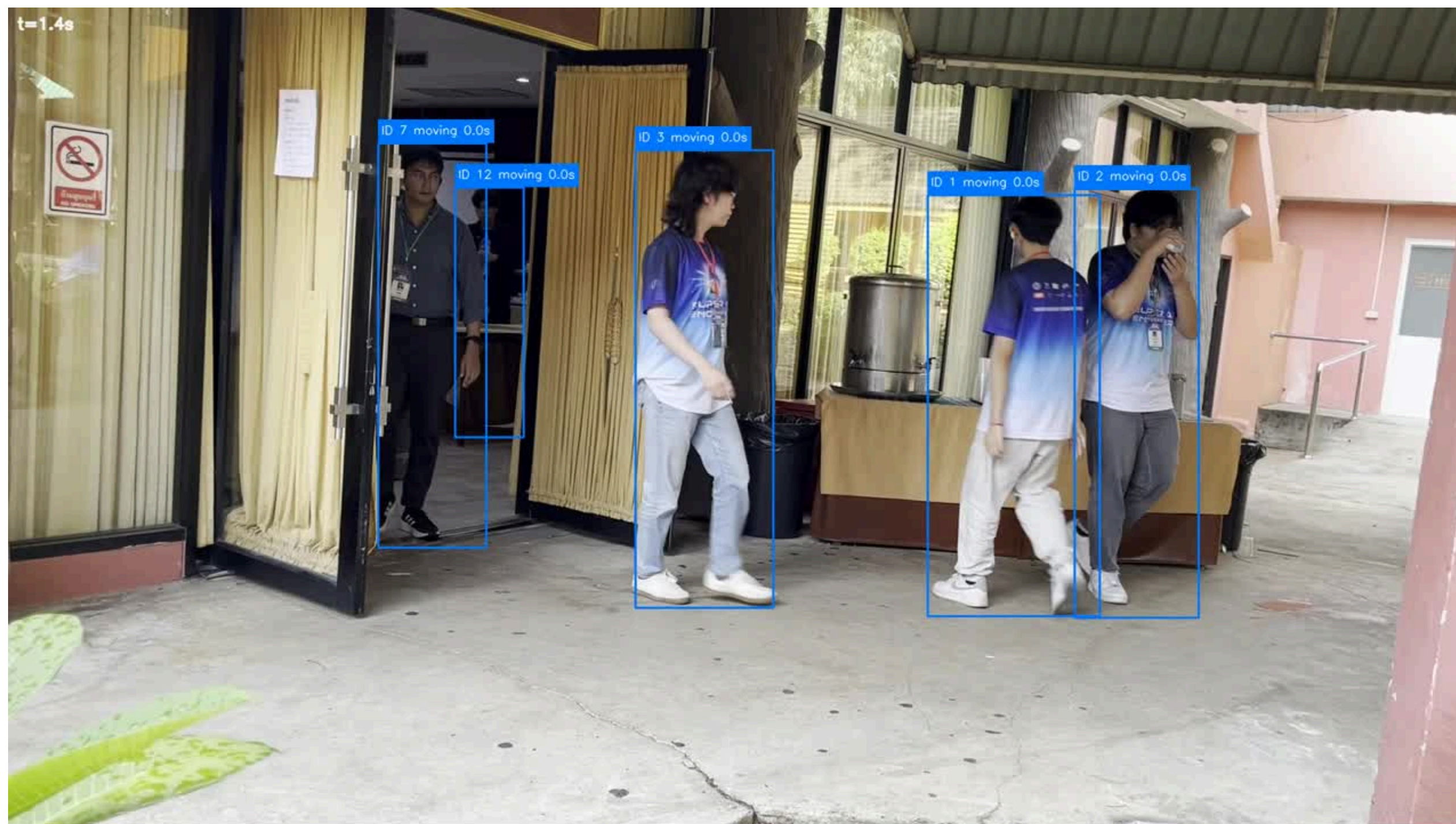
SOLUTION OF THE PROBLEM

Result Preview



SOLUTION OF THE PROBLEM

Final Answer



1920x1080 | 29.883 FPS | 2556 frames | 85.535 seconds

Winner:
track_id 161
Longest stationary
duration:
41.094 seconds
Interval:
33.699s - 74.793s

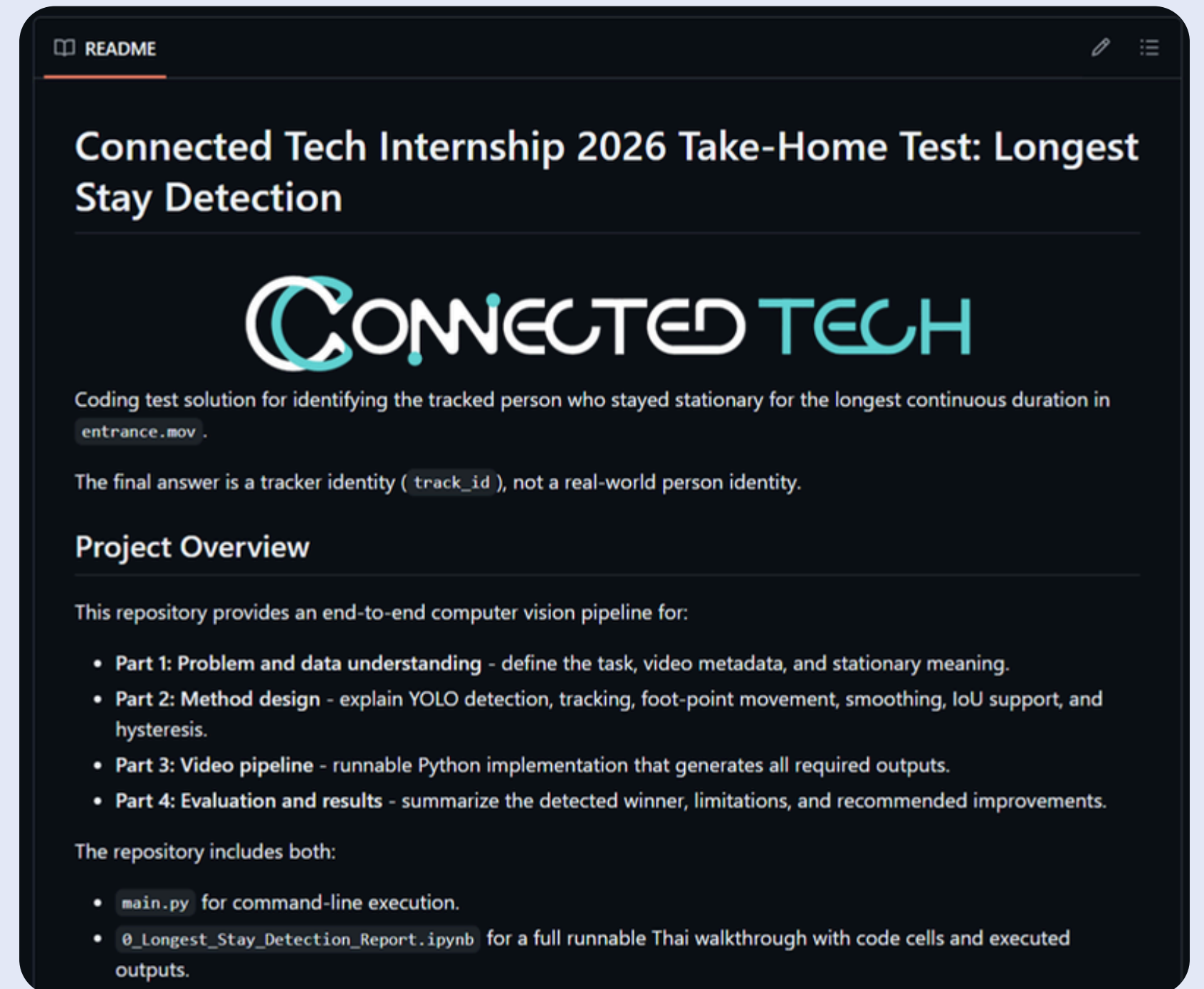
DELIVERABLES AND REPOSITORY

DELIVERABLES AND REPOSITORY

GitHub Repositories

- README.md (Example shown in the figure on the right)
- requirements.txt
- 0_Longest_Stay_Detection_Report.ipynb
(Recommended for a detailed code walkthrough)
- results folder that included
 - annotated_entrance.mp4
 - summary.json
 - tracks.csv

GitHub Repository



LIMITATIONS AND NEXT STEPS

LIMITATIONS AND NEXT STEPS

- ID switch during occlusion
- Detector jitter
- Perspective differences
- Camera shake
- Compare [yolo11s/imgsz960](#) and [bytetrack.yaml](#)
- Add global motion compensation if needed

THANK YOU.